

Village Midwife Program Analysis: Data Construction and Event Study

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2026

This document presents code from my ongoing project on the Indonesia Village Midwife (*Bidan di Desa*) program, using the Indonesia Family Life Survey (IFLS). It constructs community-level measures of village-midwife arrival from the SAR (Service Availability Roster) and PKK (Village Administration) modules, builds an individual-level analysis frame, and runs the staggered-DiD event-study specifications used in the analysis (Sun–Abraham and Callaway–Sant’Anna estimators).

Note: To keep the coding sample compact while remaining broadly representative, I use a simplified analysis sample constructed directly from IFLS Wave 5 (1989–1999 cohorts living in their birth household), without the multi-wave non-migrant restriction or the controls and survey weights used in the production specifications.

1. Setup

```
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(haven, dplyr, tidyr, purrr, ggplot2, fixest, did, broom,
               modelsummary, kableExtra)

# === EDIT HERE ===
# Path to the IFLS raw data root (containing W2/, W3/, W4/, W5/ subfolders).
# IFLS data is freely available from RAND:
# https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.html
data_path <- "/Users/wwz/Academic/Thesis_Data_IFLS/old/IFLS"

load_dta <- function(filename) {
  read_dta(file.path(data_path, filename))
}

# SAR (Service Availability Roster) and PKK (Village Administration) by wave.
# IFLS-2 (1997), IFLS-3 (2000), IFLS-4 (2007), IFLS-5 (2014/2015).
sar_14 <- load_dta("W5/cf14_all_dta/sar.dta")
pkk_14 <- load_dta("W5/cf14_all_dta/pkk.dta")

sar_07 <- load_dta("W4/cf07_all_dta/sar.dta")
pkk_07 <- load_dta("W4/cf07_all_dta/pkk.dta")

pkk_00 <- load_dta("W3/cf00_all_dta/pkk.dta") # SAR-W3 schema is not comparable

sar_97 <- load_dta("W2/cf97dta/sar_v12.dta")
pkk_97 <- load_dta("W2/cf97dta/pkk_v12.dta")
```

2. Helper Functions

I define three small functions, one per schema:

- `sar_modern_rollout()` handles the IFLS-4 and IFLS-5 SAR (x14b1, x12yr, x12b, x07, x08yr, x08b).
- `sar_old_rollout()` handles the IFLS-2 SAR (factpint, calyropn, calyrend, operates).
- `pkk_wave_rollout()` handles all four PKK waves (j10a, j10c).

Each returns a community-level data frame with the start year of the village midwife and a presence flag.

```
sar_modern_rollout <- function(sar, commid_col, survey_year) {
  sar %>%
    filter(x14b1 == 8, !is.na(.data[[commid_col]])) %>%   # 8 = village midwife
    mutate(
      open_year_fac = coalesce(
        as.integer(x12yr),
        if_else(!is.na(x12b), survey_year - as.integer(x12b), NA_integer_)
      ),
      close_year_fac = case_when(
        x07 == 1 ~ NA_integer_,                               # still operating
        !is.na(x08yr) ~ as.integer(x08yr),
        !is.na(x08b) ~ survey_year - as.integer(x08b),
        TRUE ~ NA_integer_
      ),
      open_year_fac = if_else(open_year_fac >= 1989 &
                             open_year_fac <= survey_year,
                             open_year_fac, NA_integer_)
    ) %>%
    group_by(.data[[commid_col]]) %>%
    summarise(
      start_sar = suppressWarnings(min(open_year_fac, na.rm = TRUE)),
      present_sar = as.integer(any(
        x07 == 1 |
        (!is.na(open_year_fac) & open_year_fac <= survey_year &
         (is.na(close_year_fac) | close_year_fac >= survey_year))
      )),
      n_fac = n(),
      .groups = "drop"
    ) %>%
    mutate(start_sar = if_else(is.infinite(start_sar), NA_real_, start_sar))
}
```

```
sar_old_rollout <- function(sar, commid_col = "commid97", survey_year = 1997L) {
  sar %>%
    filter(factpint == 8, !is.na(.data[[commid_col]])) %>%
    mutate(
      open_year_fac = as.integer(calyropn),
      close_year_fac = as.integer(calyrend),
      open_year_fac = if_else(open_year_fac >= 1989 &
                             open_year_fac <= survey_year,
                             open_year_fac, NA_integer_)
    ) %>%
    group_by(.data[[commid_col]]) %>%
    summarise(
      start_sar = suppressWarnings(min(open_year_fac, na.rm = TRUE)),
      present_sar = as.integer(any(
```

```

      operates == 1 |
      (!is.na(open_year_fac) & open_year_fac <= survey_year &
       (is.na(close_year_fac) | close_year_fac >= survey_year))
    )),
    n_fac = n(),
    .groups = "drop"
  ) %>%
  mutate(start_sar = if_else(is.infinite(start_sar), NA_real_, start_sar))
}

pkk_wave_rollout <- function(pkk, commid_col, survey_year) {
  pkk %>%
    filter(!is.na(.data[[commid_col]])) %>%
    mutate(
      midwife_present = as.integer(!is.na(j10a) & j10a == 1),
      start_pkk = if_else(
        midwife_present == 1 &
        as.integer(j10c) >= 1989 &
        as.integer(j10c) <= survey_year,
        as.numeric(j10c), NA_real_
      )
    ) %>%
    group_by(.data[[commid_col]]) %>%
    summarise(
      start_pkk = suppressWarnings(min(start_pkk, na.rm = TRUE)),
      present_pkk = max(midwife_present, na.rm = TRUE),
      .groups = "drop"
    ) %>%
    mutate(start_pkk = if_else(is.infinite(start_pkk), NA_real_, start_pkk))
}

```

3. SAR Rollout by Wave

```

sar_14_fac <- sar_modern_rollout(sar_14, "commid14", survey_year = 2015L) %>%
  rename(start_sar_14 = start_sar, present_sar_14 = present_sar)

sar_07_fac <- sar_modern_rollout(sar_07, "commid07", survey_year = 2007L) %>%
  rename(start_sar_07 = start_sar, present_sar_07 = present_sar)

sar_97_fac <- sar_old_rollout(sar_97, "commid97", survey_year = 1997L) %>%
  rename(start_sar_97 = start_sar, present_sar_97 = present_sar)

```

Diagnostic: how many facility-level observations report a closure year in the 2014 SAR?

```

sar_14 %>%
  filter(x14b1 == 8, !is.na(x08yr)) %>%
  summarise(n_closed = n())

```

```

## # A tibble: 1 x 1
##   n_closed
##   <int>
## 1       0

```

4. PKK Rollout by Wave

```
pkk_14_fac <- pkk_wave_rollout(pkk_14, "commid14", 2015L) %>%
  rename(start_pkk_14 = start_pkk, present_pkk_14 = present_pkk)

pkk_07_fac <- pkk_wave_rollout(pkk_07, "commid07", 2007L) %>%
  rename(start_pkk_07 = start_pkk, present_pkk_07 = present_pkk)

pkk_00_fac <- pkk_wave_rollout(pkk_00, "commid00", 2000L) %>%
  rename(start_pkk_00 = start_pkk, present_pkk_00 = present_pkk)

pkk_97_fac <- pkk_wave_rollout(pkk_97, "commid97", 1997L) %>%
  rename(start_pkk_97 = start_pkk, present_pkk_97 = present_pkk)
```

5. Community-Level Start Year

In the production pipeline, each wave-specific community ID is anchored to the IFLS-1 community ID (commid93) using the W5 household tracking file (htrack). For brevity in this sample, I work directly at the wave-specific commid level and combine through `bind_rows` on the wave keys; the combined start year is the minimum across waves and channels, clipped to [1989, survey_year].

```
start_year_sar <- bind_rows(
  sar_14_fac %>% rename(commid = commid14, start = start_sar_14),
  sar_07_fac %>% rename(commid = commid07, start = start_sar_07),
  sar_97_fac %>% rename(commid = commid97, start = start_sar_97)
) %>%
  group_by(commid) %>%
  summarise(start_year_sar = suppressWarnings(min(start, na.rm = TRUE)),
    .groups = "drop") %>%
  mutate(start_year_sar = if_else(is.infinite(start_year_sar),
    NA_real_, start_year_sar))

start_year_pkk <- bind_rows(
  pkk_14_fac %>% rename(commid = commid14, start = start_pkk_14),
  pkk_07_fac %>% rename(commid = commid07, start = start_pkk_07),
  pkk_00_fac %>% rename(commid = commid00, start = start_pkk_00),
  pkk_97_fac %>% rename(commid = commid97, start = start_pkk_97)
) %>%
  group_by(commid) %>%
  summarise(start_year_pkk = suppressWarnings(min(start, na.rm = TRUE)),
    .groups = "drop") %>%
  mutate(start_year_pkk = if_else(is.infinite(start_year_pkk),
    NA_real_, start_year_pkk))
```

6. SAR vs PKK Comparison

I keep both channels and surface the disagreement as a first-class diagnostic rather than choosing one silently. The two channels can disagree because PKK's j10c field, especially in 1997, may include all midwife types rather than village midwives only.

```
comparison <- full_join(start_year_sar, start_year_pkk, by = "commid") %>%
  mutate(
    has_sar = as.integer(!is.na(start_year_sar)),
```

```

has_pkk = as.integer(!is.na(start_year_pkk)),
has_both = as.integer(has_sar == 1 & has_pkk == 1),
delta = start_year_pkk - start_year_sar
)

comparison %>%
  summarise(
    total_communities = n(),
    with_sar = sum(has_sar),
    with_pkk = sum(has_pkk),
    with_both = sum(has_both),
    rho_sar_pkk = cor(start_year_sar, start_year_pkk,
                      use = "complete.obs"),
    share_disagree = mean(start_year_sar != start_year_pkk, na.rm = TRUE),
    median_delta_yr = median(delta, na.rm = TRUE),
    diff_range = mean(abs(start_year_sar - start_year_pkk), na.rm = TRUE)
  )

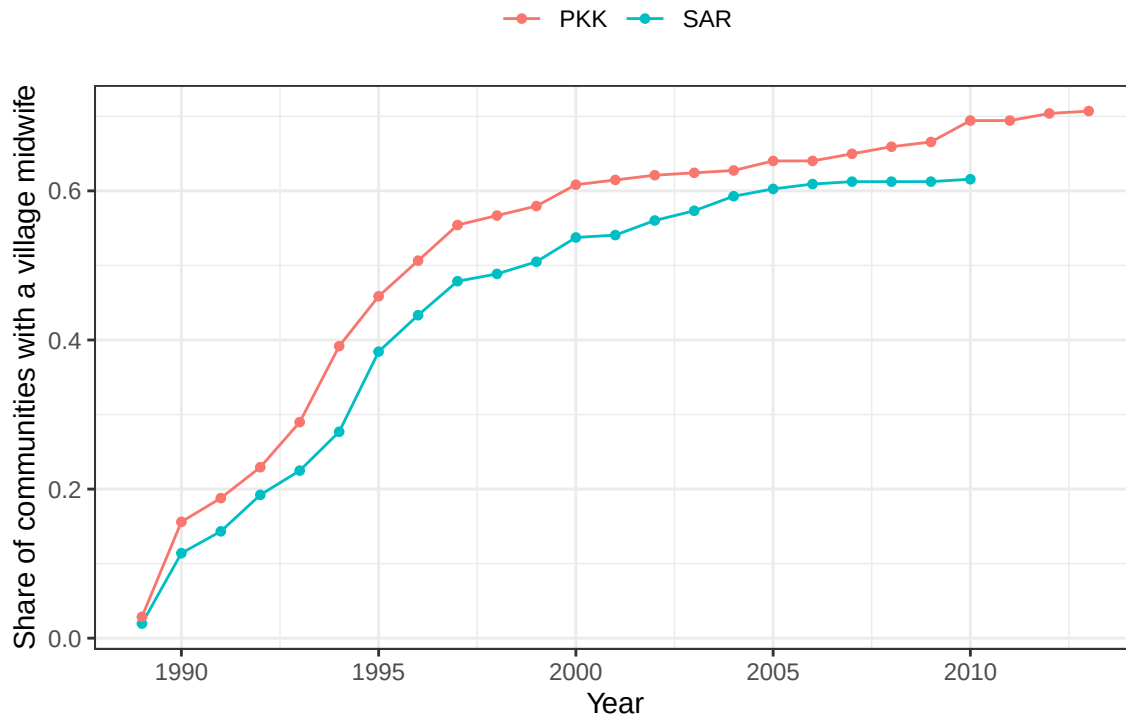
## # A tibble: 1 x 8
##   total_communities with_sar with_pkk with_both rho_sar_pkk share_disagree
##           <int>    <int>    <int>    <int>    <dbl>         <dbl>
## 1           314      189      222      152      0.0976         0.829
## # i 2 more variables: median_delta_yr <dbl>, diff_range <dbl>

rollout_curve <- function(df, col, label) {
  df %>%
    drop_na(all_of(col)) %>%
    distinct(commid, .data[[col]]) %>%
    count(.data[[col]], name = "n_new") %>%
    rename(start_year = !!col) %>%
    complete(start_year = seq(min(start_year), max(start_year)),
             fill = list(n_new = 0)) %>%
    arrange(start_year) %>%
    mutate(
      cumulative = cumsum(n_new),
      share = cumulative / nrow(df),
      source = label
    )
}

rollout_plot_df <- bind_rows(
  rollout_curve(start_year_sar, "start_year_sar", "SAR"),
  rollout_curve(start_year_pkk, "start_year_pkk", "PKK")
)

ggplot(rollout_plot_df,
  aes(x = start_year, y = share, color = source, group = source)) +
  geom_line() + geom_point(size = 1.2) +
  labs(x = "Year", y = "Share of communities with a village midwife",
       color = NULL) +
  theme_bw() + theme(legend.position = "top")

```



7. Individual-Level Analysis Frame

Treatment is exposure to a village midwife in the early-life window (in utero through age 3). At the community level, $G = \text{start_sar_14} - 1$ is the first cohort exposed in utero; never-treated communities act as the control group. Below I load the IFLS-5 individual roster (`bk_ar1`) and adult anthropometrics (`bus_us`) to build a child-level frame, then merge in community-level treatment timing. The outcome is adult height (`us04`), a standard early-life human-capital marker. Cohorts are restricted to 1989–1999 (children plausibly exposed in utero or during their first three years).

```
roster_w5 <- load_dta("W5/hh14_all_dta/bk_ar1.dta")
height_w5 <- load_dta("W5/hh14_all_dta/bus_us.dta")
htrack_w5 <- load_dta("W5/hh14_all_dta/htrack.dta")

# hhid14 -> commid14 mapping.
hh_to_commid <- htrack_w5 %>%
  transmute(hhid14 = as.character(hhid14),
            commid14 = as.character(commid14)) %>%
  distinct()

# Individual frame: pidlink, sex, birth_year, commid14.
individuals <- roster_w5 %>%
  transmute(
    pidlink = as.character(pidlink),
    hhid14 = as.character(hhid14),
    sex = as.integer(ar07), # 1 = male, 3 = female
    birth_year = as.integer(ar08yr)
  ) %>%
  filter(birth_year >= 1989, birth_year <= 1999) %>% # treatable cohorts
  left_join(hh_to_commid, by = "hhid14")

# Adult height in cm, with sanity bounds.
```

```

heights <- height_w5 %>%
  transmute(pidlink = as.character(pidlink),
            height_cm = as.numeric(us04)) %>%
  mutate(height_cm = if_else(height_cm >= 100 & height_cm <= 210,
                             height_cm, NA_real_))

# Merge on individual + community-level treatment.
analysis <- individuals %>%
  left_join(heights, by = "pidlink") %>%
  left_join(
    sar_14_fac %>% transmute(commid14 = as.character(commid14),
                             start_sar_14),
    by = "commid14"
  ) %>%
  mutate(
    G = if_else(!is.na(start_sar_14),
                as.integer(start_sar_14) - 1L, NA_integer_),
    first_treat = G, # NA = never-treated
    event_time = if_else(!is.na(G), birth_year - G, NA_integer_)
  ) %>%
  filter(!is.na(height_cm), !is.na(commid14))

analysis %>%
  summarise(
    n_individuals = n(),
    n_communities = n_distinct(commid14),
    n_treated_comm = n_distinct(commid14[!is.na(G)]),
    n_never_treated = n_distinct(commid14[is.na(G)]),
    mean_height_cm = mean(height_cm),
    cohort_min = min(birth_year),
    cohort_max = max(birth_year)
  )

## # A tibble: 1 x 7
##   n_individuals n_communities n_treated_comm n_never_treated mean_height_cm
##           <int>         <int>         <int>         <int>         <dbl>
## 1       10470         1385           74          1311          158.
## # i 2 more variables: cohort_min <int>, cohort_max <int>

```

8. Descriptive Statistics

I report two summaries of the analysis frame: an overall descriptive table and a balance comparison between treated and never-treated communities. Modest pre-treatment imbalances are expected because rollout was not random; the staggered-DiD estimators in the next sections account for this through community fixed effects and a never-treated control group.

```

desc_data <- analysis %>%
  transmute(
    `Adult height (cm)` = height_cm,
    `Birth year` = birth_year,
    `Female` = as.integer(sex == 3),
    `Treated community` = as.integer(!is.na(G)),
    `Exposure (in utero - age 3)` = as.integer(!is.na(G) &
                                                birth_year >= G &

```

Table 1: Summary statistics, analysis sample

	N	Mean	SD	Min	Median	Max
Adult height (cm)	10470	157.71	8.31	104.60	157.10	186.40
Birth year	10470	1993.94	3.23	1989.00	1994.00	1999.00
Female	10470	0.55	0.50	0.00	1.00	1.00
Treated community	10470	0.16	0.37	0.00	0.00	1.00
Exposure (in utero - age 3)	10470	0.03	0.17	0.00	0.00	1.00

Table 2: Treated vs never-treated communities

	Never-treated (N=8764)		Treated (N=1706)		Diff. in Means	Std. Error
	Mean	Std. Dev.	Mean	Std. Dev.		
Adult height (cm)	157.8	8.3	157.4	8.3	-0.4	0.2
Birth year	1993.9	3.2	1993.9	3.2	0.0	0.1
Female	0.5	0.5	0.6	0.5	0.0	0.0
Exposure (in utero - age 3)	0.0	0.0	0.2	0.4	0.2	0.0

```

    birth_year <= G + 3L)
  )

datasummary(All(desc_data) ~ N + Mean + SD + Min + Median + Max,
  data = desc_data,
  title = "Summary statistics, analysis sample",
  output = "kableExtra")

desc_data %>%
  mutate(group = if_else(`Treated community` == 1,
    "Treated", "Never-treated")) %>%
  select(-`Treated community`) %>%
  datasummary_balance(~ group, data = .,
    title = "Treated vs never-treated communities",
    output = "kableExtra")

```

9. Sun–Abraham Event Study on Adult Height

```

es_fit <- feols(
  height_cm ~ sunab(first_treat, birth_year) | commid14 + birth_year,
  data = analysis,
  cluster = ~ commid14
)

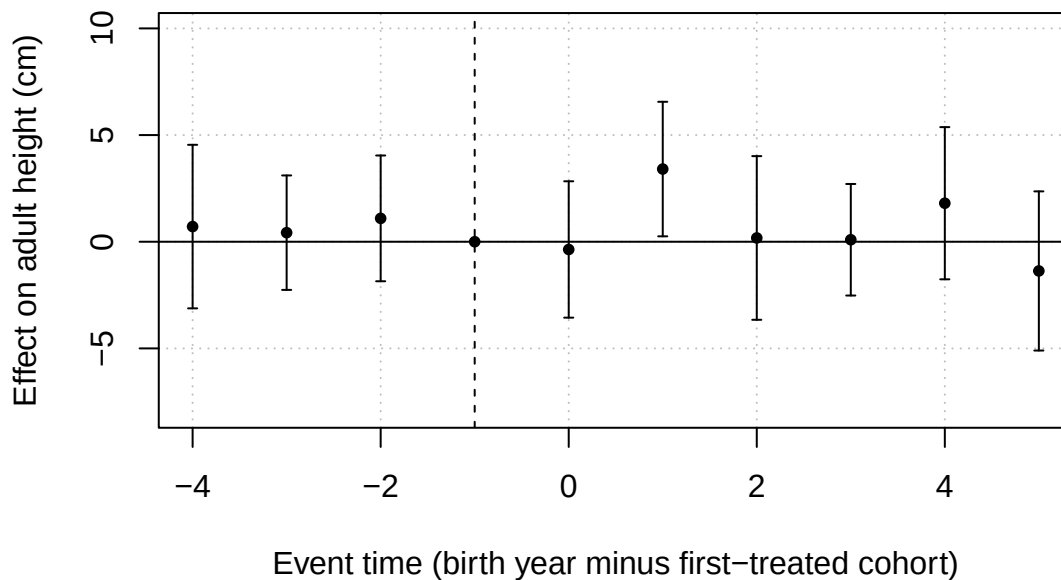
summary(es_fit, agg = "ATT")

## OLS estimation, Dep. Var.: height_cm
## Observations: 1,704
## Fixed-effects: commid14: 73, birth_year: 11
## Standard-errors: Clustered (commid14)
##      Estimate Std. Error   t value Pr(>|t|)

```

```
## ATT -0.057247    1.19412 -0.047941    0.9619
## ... 2 variables were removed because of collinearity
## (birth_year::0:cohort::1998 and birth_year::8:cohort::1991)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 7.64125    Adj. R2: 0.077213
##                Within R2: 0.050457
```

```
iplot(es_fit,
      xlim    = c(-4, 5),
      ylim    = c(-8, 10),
      ref.line = -1,
      xlab     = "Event time (birth year minus first-treated cohort)",
      ylab     = "Effect on adult height (cm)",
      main     = "")
```



10. Callaway–Sant’Anna ATT (event-times 0–3)

```
cs_data <- analysis %>%
  mutate(
    G_cs      = if_else(!is.na(G), as.integer(G), 0L),
    commid_num = as.integer(factor(commid14))
  ) %>%
  as.data.frame()

cs_att <- did::att_gt(
  yname = "height_cm", tname = "birth_year", idname = "commid_num",
  gname = "G_cs", data = cs_data, control_group = "nevertreated",
  bstrap = FALSE, cband = FALSE, panel = FALSE,
  allow_unbalanced_panel = TRUE
)

cs_dyn <- did::aggte(cs_att, type = "dynamic", min_e = 0, max_e = 3,
  na.rm = TRUE)
```

```

tibble(
  estimator = "Callaway--Sant'Anna",
  outcome   = "Adult height (cm)",
  att_0_to_3 = cs_dyn$overall.att,
  se        = cs_dyn$overall.se
)

## # A tibble: 1 x 4
##   estimator      outcome      att_0_to_3    se
##   <chr>          <chr>          <dbl> <dbl>
## 1 Callaway--Sant'Anna Adult height (cm) -0.0448  1.16

# Full event-study aggregation, displayed over the same window as SA.
cs_dyn_full <- did::aggte(cs_att, type = "dynamic",
  min_e = -4, max_e = 5, na.rm = TRUE)

did::ggdid(cs_dyn_full) +
  labs(x = "Event time (birth year minus first-treated cohort)",
    y = "Effect on adult height (cm)",
    title = NULL) +
  theme_bw()

```

